Chapter 1: Function Transformations

MULTIPLE CHOICE

1. Compared to the graph of the base function $f(x) = x^2$, the graph of the function $g(x) = x^2 - 9$ is translated

A 9 units to the left
B 9 units to the right
C 9 units down
D 9 units up

ANS: C DIF: Easy OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations KEY: vertical translation

2. Compared to the graph of the base function $f(x) = |x|$, the graph of the function $g(x) + 5 = |x|$ is translated

A 5 units to the right
B 5 units up
C 5 units down
D 5 units to the left

ANS: C DIF: Average OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations KEY: vertical translation

3. Compared to the graph of the base function $f(x) = x^2$, the graph of the function $g(x) = (x - 4)^2$ is translated

A 4 units to the right
B 4 units to the left
C 4 units down
D 4 units up

ANS: A DIF: Easy OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations KEY: horizontal translation

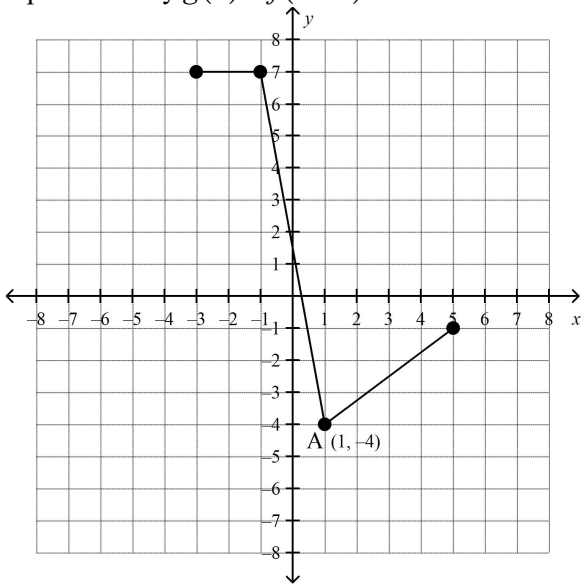
4. Compared to the graph of the base function $f(x) = |x|$, the graph of the function $g(x) = |x + 9|$ is translated

A 9 units to the right
B 9 units up
C 9 units down
D 9 units to the left

ANS: D DIF: Easy OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations KEY: horizontal translation

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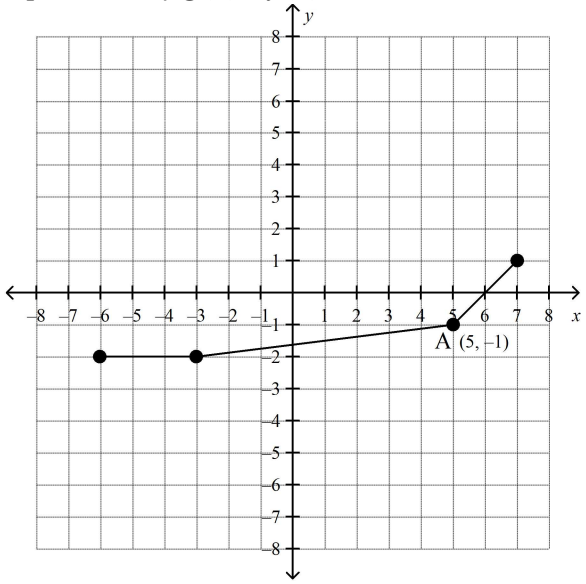
5. Given the graph of $f(x)$ shown below, what are the coordinates of point A if the transformed graph is represented by $g(x) = f(x-2)$?



A $(3, -4)$
B $(-1, -4)$
C $(-1, -2)$
D $(3, -6)$

ANS: A DIF: Average OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations KEY: horizontal translation

6. Given the graph of $f(x)$ shown below, what are the coordinates of point A if the transformed graph is represented by $g(x) = f(x) + 3$?

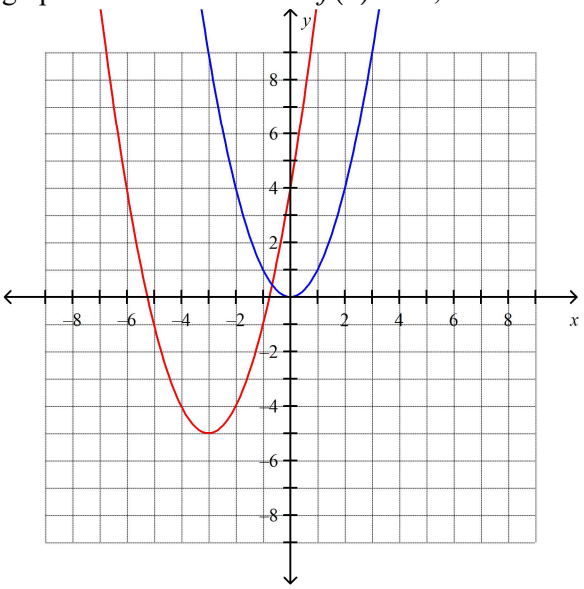


A $(5, 2)$
B $(2, -4)$
C $(2, -1)$
D $(8, 2)$

ANS: A DIF: Average OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations KEY: vertical translation

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7. What is the equation of the transformed function, $g(x)$, after the transformations are applied to the graph of the base function $f(x) = x^2$, shown in blue, to obtain the graph of $g(x)$, shown in red?



A $g(x) + 3 = (x - 5)^2$

B $g(x) = (x + 3)^2 - 5$

C $g(x) - 5 = (x + 3)^2$

D $g(x) = (x - 5)^2 + 3$

ANS: B DIF: Average OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations
KEY: horizontal translation | vertical translation

8. The function $f(x) = -g(x)$ represents a transformation that can best be described as

A a reflection in the x-axis

B a reflection in the x-axis and the y-axis

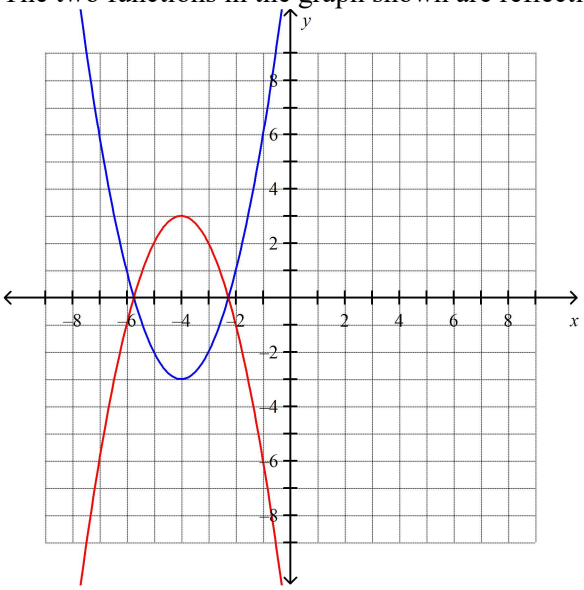
C a reflection in the y-axis

D a reflection in the line $y = x$

ANS: A DIF: Easy OBJ: Section 1.2 NAT: RF5
TOP: Reflections and Stretches KEY: reflection

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9. The two functions in the graph shown are reflections of each other. Select the type of reflection(s).



A a reflection in the line $y = x$
B a reflection in the x-axis and the y-axis
C a reflection in the y-axis
D a reflection in the x-axis

ANS: D DIF: Average OBJ: Section 1.2 NAT: RF5
TOP: Reflections and Stretches KEY: reflection

10. When a function is reflected in the x-axis, the coordinates of point (x, y) become
A $(x, -y)$
B $(-x, y)$
C $(-x, -y)$
D (x, y)

ANS: A DIF: Easy OBJ: Section 1.1 NAT: RF5
TOP: Reflections and Stretches KEY: reflection

11. When $a > 0$, the function $g(x) = ax^2$ has what relationship to the base function $f(x) = x^2$?
A $f(x)$ is stretched vertically by a factor of |a| and reflected in the x-axis
B $f(x)$ is stretched horizontally by a factor of 1/|a|
C $f(x)$ is stretched vertically by a factor of |a|
D $f(x)$ is stretched horizontally by a factor of 1/|a| and reflected in the y-axis

ANS: C DIF: Easy OBJ: Section 1.2 NAT: RF3 | RF5
TOP: Reflections and Stretches KEY: vertical stretch | reflection

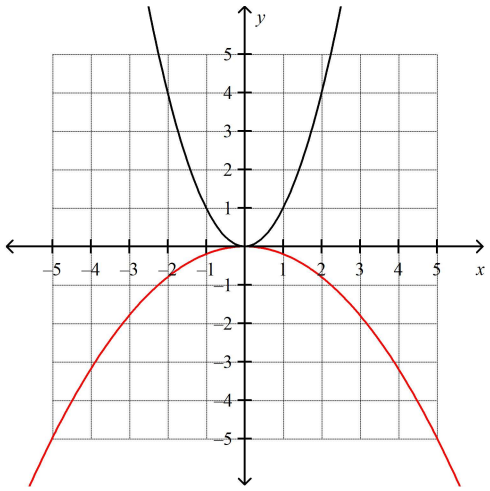
12. When $b > 0$, the function $g(x) = |bx|$ has what relationship to the base function $f(x) = |x|$?
A $f(x)$ is stretched vertically by a factor of |b| and reflected in the x-axis
B $f(x)$ is stretched vertically by a factor of |b|
C $f(x)$ is stretched horizontally by a factor of 1/|b| and reflected in the y-axis
D $f(x)$ is stretched horizontally by a factor of 1/|b|

ANS: D DIF: Easy OBJ: Section 1.2 NAT: RF3 | RF5
TOP: Reflections and Stretches KEY: horizontal stretch | reflection

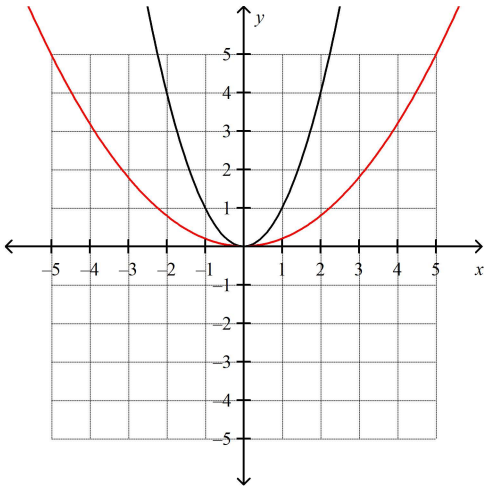
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13. Which of the graphs shown below represents the base function $f(x) = x^2$ and the stretched function $g(x) = -\frac{1}{5}x^2$?

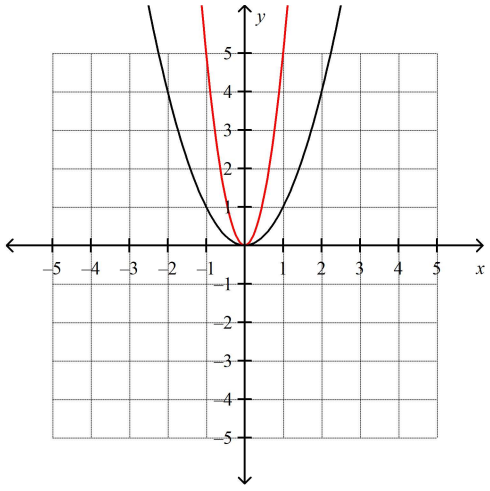
A



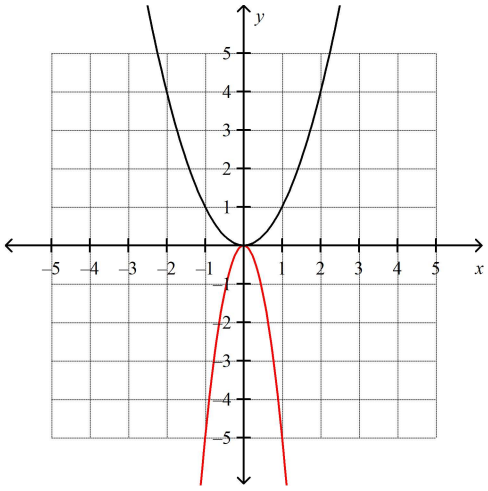
C



B



D

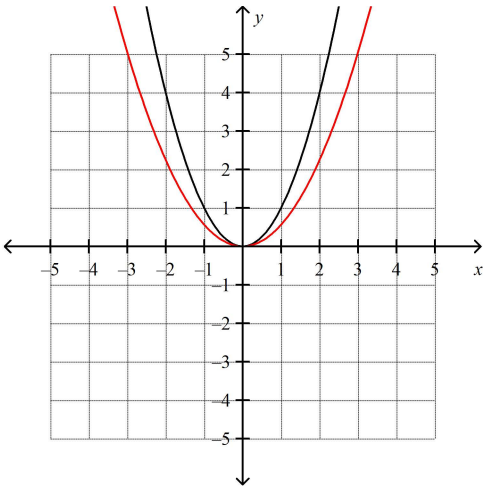


ANS: A DIF: Average OBJ: Section 1.2 NAT: RF3 | RF5
TOP: Reflections and Stretches KEY: graph | vertical stretch | reflection

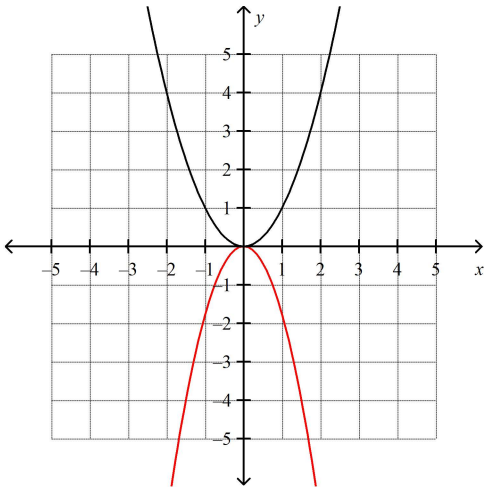
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14. Which of the graphs shown below represents the base function $f(x) = x^2$ and the stretched function $g(x) = (\frac{3}{4}x)^2$?

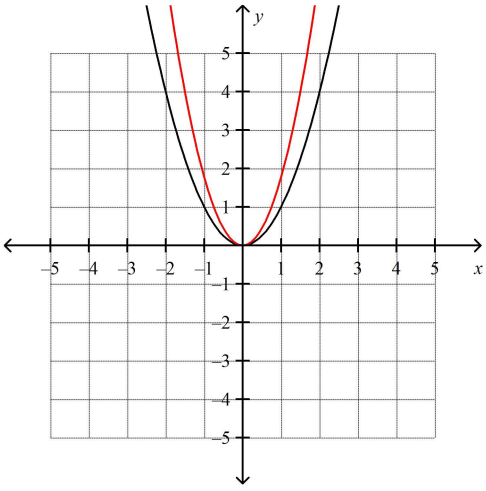
A



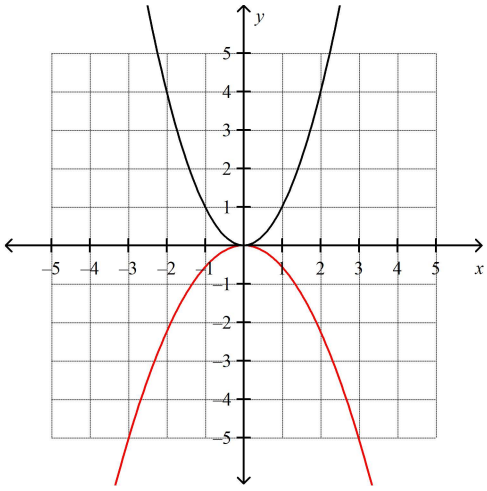
C



B



D



ANS: A DIF: Average OBJ: Section 1.2 NAT: RF3 | RF5
TOP: Reflections and Stretches KEY: graph | horizontal stretch | reflection

15. What are the coordinates of the invariant point(s) when the function $y = |x| - 2$ is reflected in the y-axis?

A (2, –2)
B (–2, 0) and (2, 0)
C (0, –2)
D (0, 2)

ANS: C DIF: Average OBJ: Section 1.2 NAT: RF3
TOP: Reflections and Stretches KEY: invariant point

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16. When the value of a is less than –1, the function $g(x) = ax^2$ has what relationship to the base function $f(x) = x^2$?

A $f(x)$ is compressed vertically
B $f(x)$ is reflected and compressed vertically
C $f(x)$ is stretched vertically
D $f(x)$ is reflected and stretched vertically

ANS: D DIF: Average OBJ: Section 1.2 NAT: RF3 | RF5
TOP: Reflections and Stretches KEY: reflection | vertical stretch

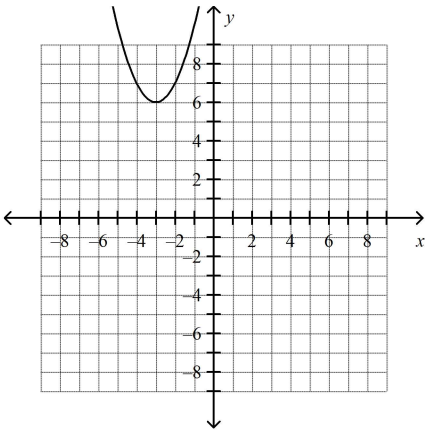
17. Compared to the graph of the base function $f(x) = x^2$, the graph of the function $g(x) = (x-4)^2 + 9$ is translated

A 9 units to the left and 4 units down
B 4 units to the left and 9 units down
C 9 units to the right and 4 units up
D 4 units to the right and 9 units up

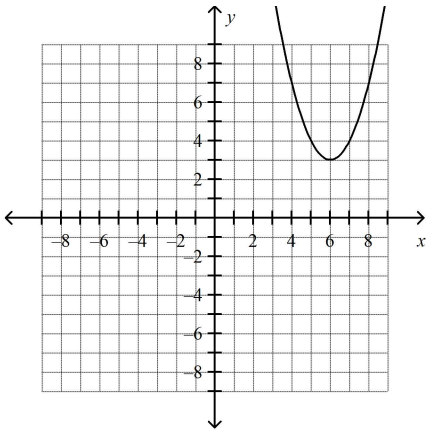
ANS: D DIF: Easy OBJ: Section 1.3 NAT: RF4
TOP: Combining Transformations KEY: horizontal translation | vertical translation

18. Which is the graph of the function $f(x) = (x-6)^2 + 3$?

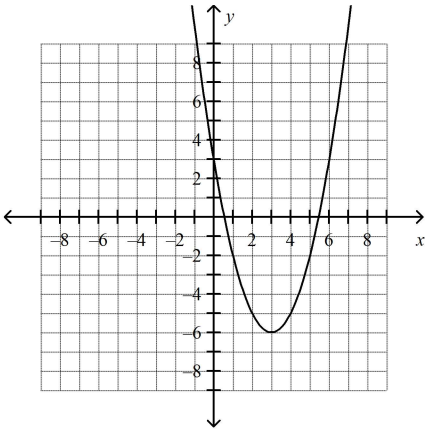
A



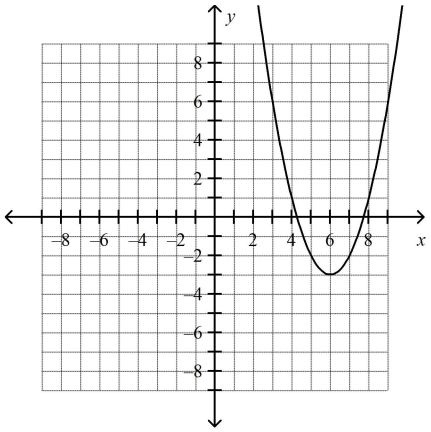
C



B



D



ANS: C DIF: Easy OBJ: Section 1.3 NAT: RF4
TOP: Combining Transformations KEY: graph | horizontal translation | vertical translation

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19. Which choice best describes the combination of transformations that must be applied to the graph of $f(x) = |x|$ to obtain the graph of $g(x) = f(2x - 4)$?

A a horizontal stretch by a factor of 2 and a horizontal translation of 2 units to the left

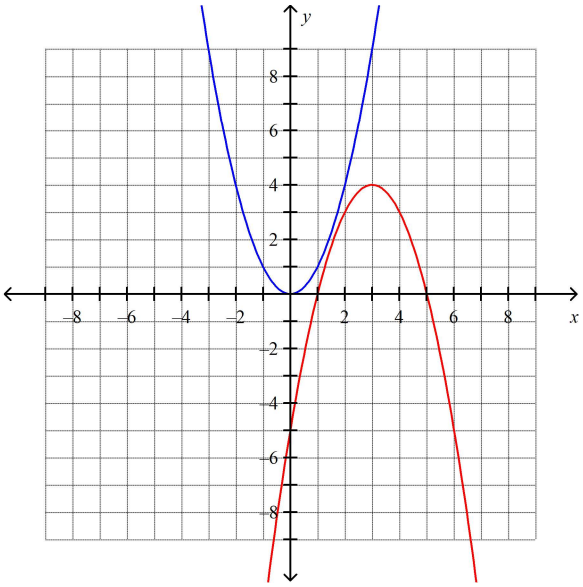
B a horizontal stretch by a factor of $\frac{1}{2}$ and a horizontal translation of 4 units to the right

C a horizontal stretch by a factor of $\frac{1}{2}$ and a horizontal translation of 2 units to the right

D a horizontal stretch by a factor of –2 and a horizontal translation of 2 units to the right

ANS: C DIF: Difficult OBJ: Section 1.3 NAT: RF4
TOP: Combining Transformations KEY: horizontal stretch | horizontal translation

20. In the following graph, which transformations must be applied to the blue curve to obtain the red curve?



A a reflection in the *x*-axis, a vertical translation of 4 units up, and a horizontal translation of 3 units to the right

B a reflection in the *x*-axis, a vertical translation of 3 units up, and a horizontal translation of 4 units to the right

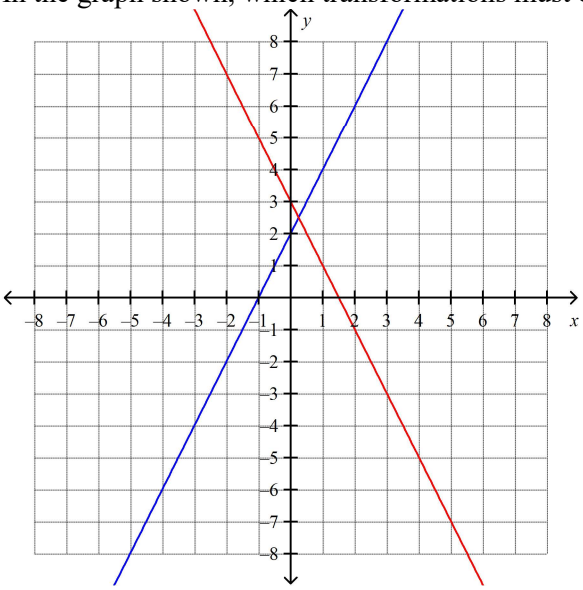
C a reflection in the *x*-axis, a vertical translation of 3 units down, and a horizontal translation of 4 units to the right

D a reflection in the *x*-axis, a vertical translation of 4 units up, and a horizontal translation of 3 units to the left

ANS: A DIF: Average OBJ: Section 1.3 NAT: RF4 | RF5
TOP: Combining Transformations
KEY: graph | horizontal translation | vertical translation | reflection

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21. In the graph shown, which transformations must be applied to the blue curve to obtain the red curve?



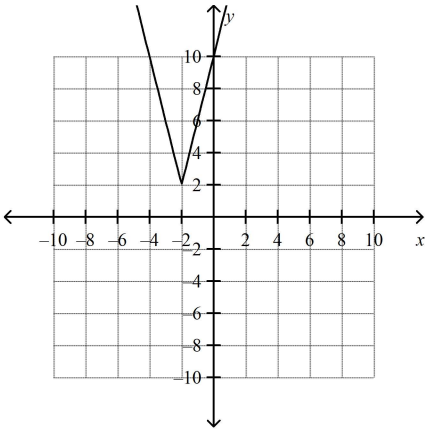
A a reflection in the *x*-axis and a translation of 5 units down
B a reflection in the *y*-axis and a translation of 5 units up
C a reflection in the *x*-axis and a translation of 5 units up
D a reflection in the *y*-axis and a translation of 5 units down

ANS: C DIF: Average OBJ: Section 1.3 NAT: RF4
TOP: Combining Transformations KEY: graph | vertical translation | reflection

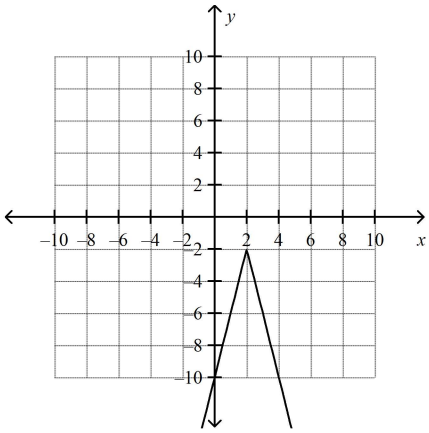
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22. Which of the following graphs represents the graph of the function $f(x) = |x|$ transformed to $f(x) = 2|-2x + 4| + 2$?

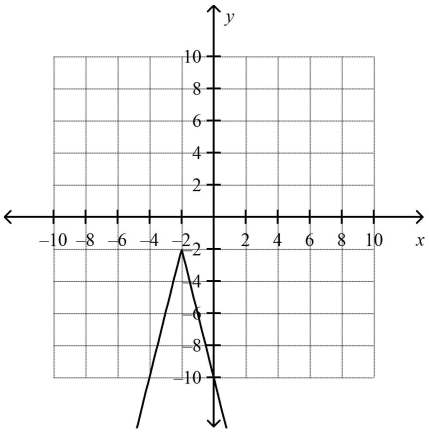
A



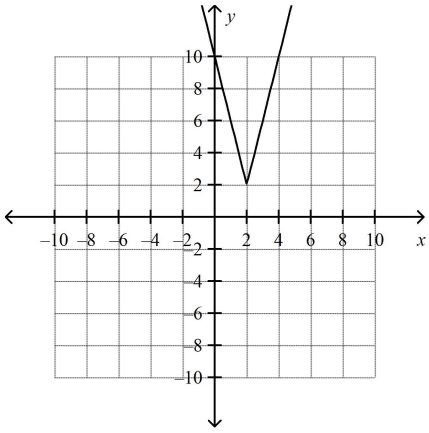
C



B



D

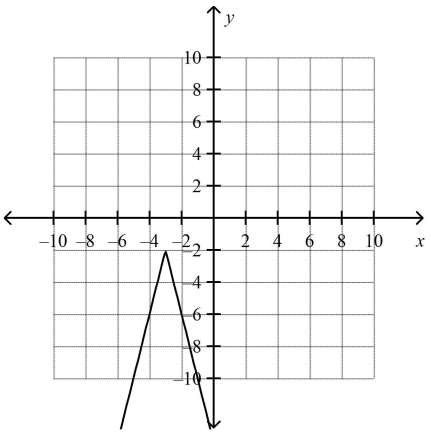


ANS: D DIF: Difficult OBJ: Section 1.3 NAT: RF4 | RF5
TOP: Combining Transformations
KEY: graph | vertical translation | horizontal translation | stretch | reflection

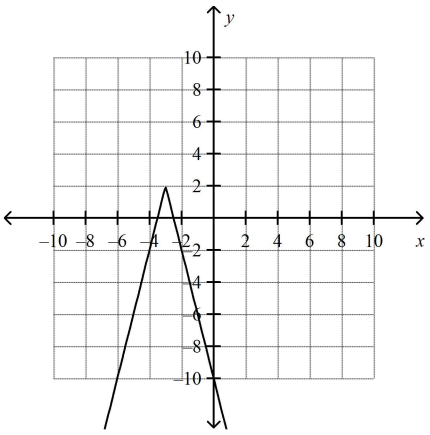
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23. When the function $f(x) = |x|$ is transformed to $f(x) = -4|x+3|+2$, the graph looks like

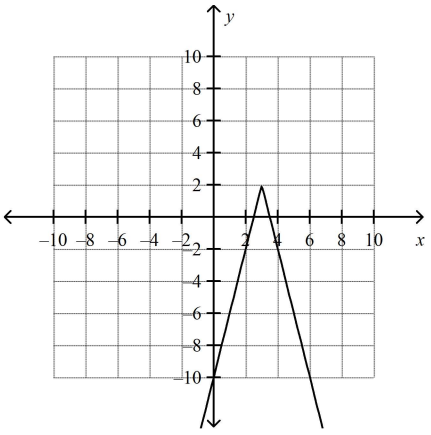
A



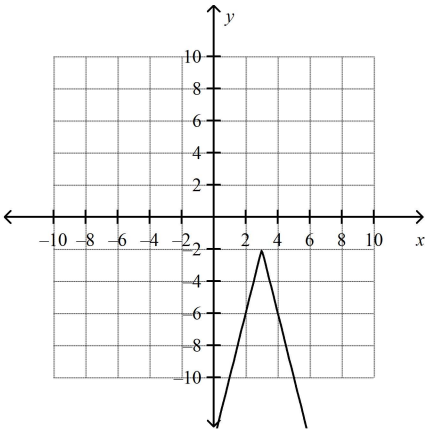
C



B



D



ANS: C DIF: Average OBJ: Section 1.3 NAT: RF4 | RF5
TOP: Combining Transformations
KEY: graph | vertical translation | horizontal translation | stretch | reflection

24. Which of the following functions is the correct inverse for the function $f(x) = 3x + 5$?

A $f^{-1}(x) = \frac{1}{3}x - \frac{5}{3}$

B $f^{-1}(x) = -\frac{1}{3}x + \frac{5}{3}$

C $f^{-1}(x) = -\frac{1}{3}x - \frac{5}{3}$

D $f^{-1}(x) = \frac{1}{3}x + \frac{5}{3}$

ANS: A DIF: Easy OBJ: Section 1.4 NAT: RF6
TOP: Inverse of a Relation KEY: inverse of a function | function notation

25. Which of the following functions is the correct inverse for the function $f(x) = -\frac{9}{2}x + 6$?

A $f^{-1}(x) = -\frac{2}{9}x + \frac{4}{3}$

B $f^{-1}(x) = \frac{9}{2}x + \frac{4}{3}$

C $f^{-1}(x) = -\frac{2}{9}x - \frac{4}{3}$

D $f^{-1}(x) = \frac{9}{2}x - \frac{4}{3}$

ANS: A DIF: Average OBJ: Section 1.4 NAT: RF6
TOP: Inverse of a Relation KEY: inverse of a function | function notation

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26. Which of the following functions is the correct inverse for the function $f(x) = x^2 + 7$, $\{x \mid x \geq 0, x \in \mathrm{R}\}$?

A $f^{-1}(x) = (x-7)^2$

B $f^{-1}(x) = \sqrt{x} + 7$

C $f^{-1}(x) = \sqrt{x-7}$

D $f^{-1}(x) = \sqrt{x+7}$

ANS: C DIF: Average OBJ: Section 1.4 NAT: RF6
TOP: Inverse of a Relation KEY: inverse of a function | function notation

27. Which of the following relations is the correct inverse for the function $f(x) = x^2 + 6x + 2$?

A $f^{-1}(x) = 3 \pm \sqrt{x-7}$

B $f^{-1}(x) = 7 \pm \sqrt{x+3}$

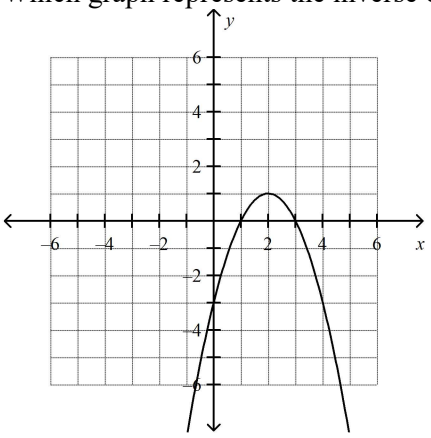
C $f^{-1}(x) = -3 \pm \sqrt{x+7}$

D $f^{-1}(x) = -7 \pm \sqrt{x-3}$

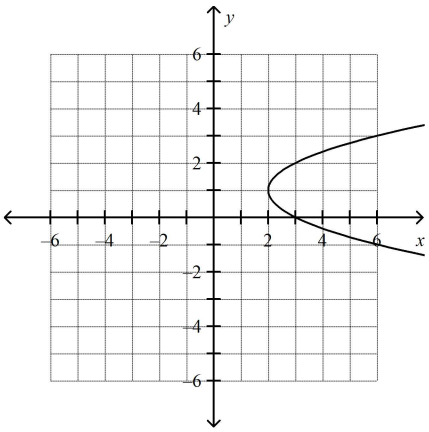
ANS: C DIF: Difficult + OBJ: Section 1.4 NAT: RF6
TOP: Inverse of a Relation
KEY: inverse of a function | function notation | completing the square

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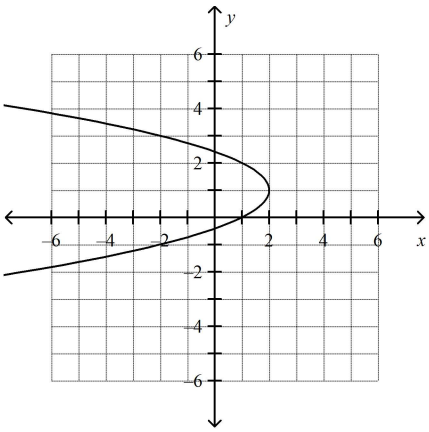
28. Which graph represents the inverse of the graph shown?



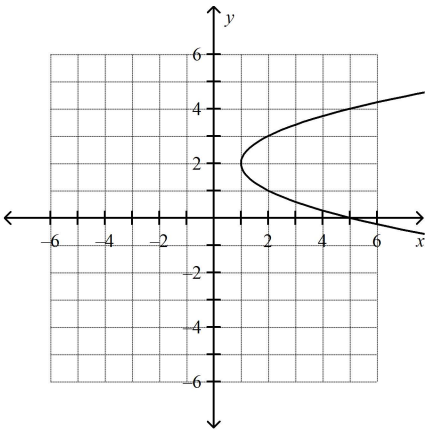
A



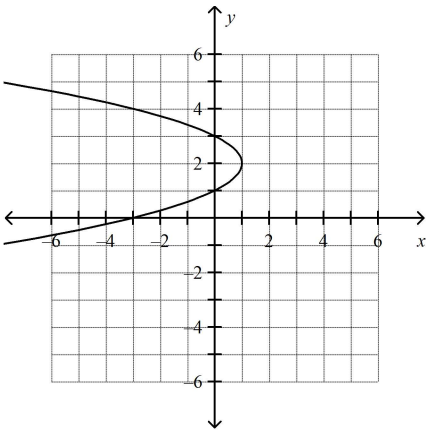
C



B



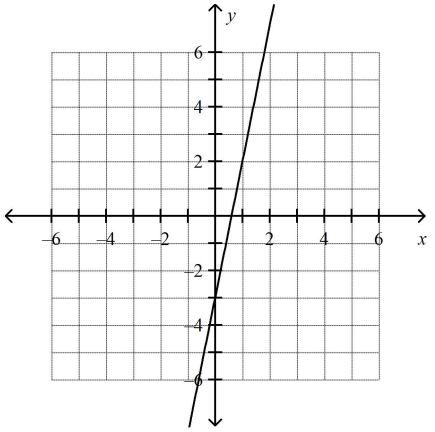
D



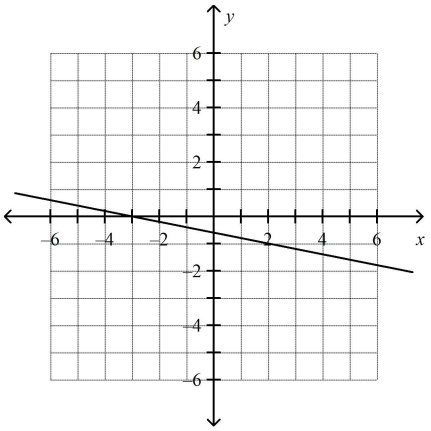
ANS: D DIF: Average OBJ: Section 1.4 NAT: RF6
TOP: Inverse of a Relation KEY: graph | inverse of a function

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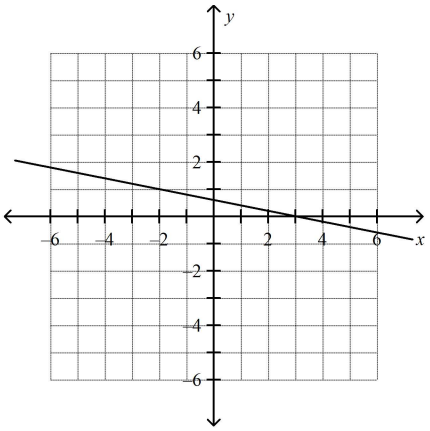
29. Which graph represents the inverse of the function shown?



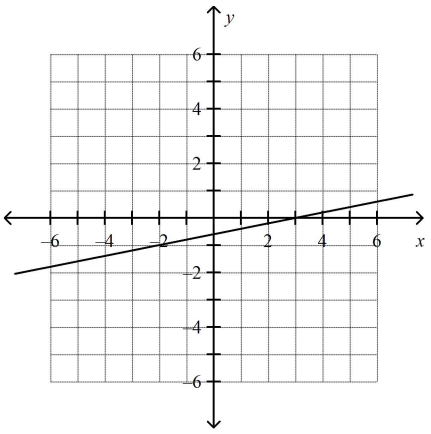
A



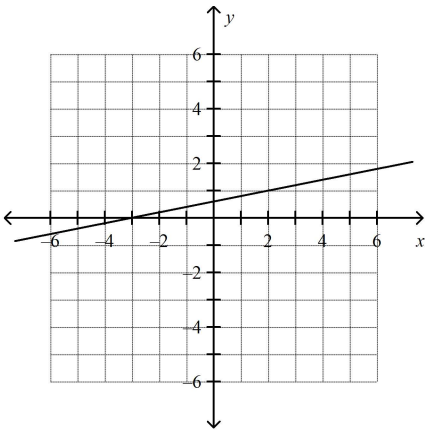
C



B



D



ANS: D DIF: Easy OBJ: Section 1.4 NAT: RF6
TOP: Inverse of a Relation KEY: graph | inverse of a function

30. The equation of the inverse of $f(x) = -9x$ is

A $f^{-1}(x) = \frac{x}{-9}$

B $f^{-1}(x) = \frac{x}{9}$

C $f^{-1}(x) = 9x$

D $f^{-1}(x) = -9x$

ANS: A DIF: Easy OBJ: Section 1.4 NAT: RF6
TOP: Inverse of a Relation KEY: inverse of a function | function notation

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COMPLETION

1. _____ are transformations that cause the graph of a function to change shape without changing the orientation of the graph.

ANS: Stretches

DIF: Easy OBJ: Section 1.1 | Section 1.2 NAT: RF2 | RF3
TOP: Horizontal and Vertical Translations | Reflections and Stretches
KEY: translation | stretch | reflection

2. The graph of $g(x) = f(x-2)$ is the graph of $f(x)$ translated 2 units _____.

ANS: to the right

DIF: Easy OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations
KEY: horizontal translation | vertical translation

3. The _____ is (are) invariant on a function after a reflection in the y-axis.

ANS: the y-intercept

DIF: Easy OBJ: Section 1.2 NAT: RF3 TOP: Reflections and Stretches
KEY: reflection | invariant point

4. The graph of $f(x)$ is reflected in the y-axis, translated 8 units to the left, and translated 7 units down. The equation of the transformed function is $g(x)$ = _____.

ANS: $f(-(x+8))-7$

DIF: Average OBJ: Section 1.3 NAT: RF4 | RF5 TOP: Combining Transformations
KEY: translation | reflection

5. The inverse of the function $f(x) = \frac{5}{4}x - 5$ is _____.

ANS: $f^{-1}(x) = \frac{4}{5}x + 4$

DIF: Average OBJ: Section 1.4 NAT: RF6 TOP: Inverse of a Relation
KEY: inverse of a function

MATCHING

Match each definition or explanation given below to its corresponding term.

A	transformation	**D**	image point
B	mapping	**E**	reflection
C	inverse of a relation	**F**	invariant point

1. the point that is the result of a transformation of the original point on the graph

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2. a point that is mapped to itself by the function

3. a rule of correspondence established between sets that associates each element of a set with an element in the same or another set

4. a relation that undoes another relation

5. a change made to a figure or graph of a relation that results in a shift or change in shape of the figure or graph

1. ANS: D DIF: Easy OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations KEY: image point

2. ANS: F DIF: Easy OBJ: Section 1.2 NAT: RF5
TOP: Reflections and Stretches KEY: invariant point

3. ANS: B DIF: Easy OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations KEY: mapping

4. ANS: C DIF: Easy OBJ: Section 1.4 NAT: RF6
TOP: Inverse of a Relation KEY: inverse of a function

5. ANS: A DIF: Easy OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations KEY: transformation

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SHORT ANSWER

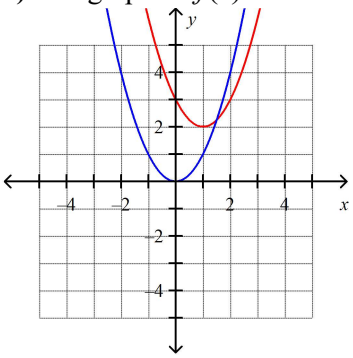
1. Create a graph of $g(x) = f(x-1)+2$ for each base function given, using transformations.

a) $f(x) = x^2$

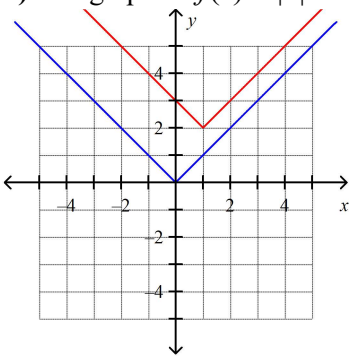
b) $f(x) = |x|$

ANS:

a) The graph of $f(x) = x^2$ is shown in blue and the graph of $g(x) = (x-1)^2 + 2$ is shown in red.



b) The graph of $f(x) = |x|$ is shown in blue and the graph of $g(x) = |x-1|+2$ is shown in red.



DIF: Average OBJ: Section 1.3 NAT: RF4 TOP: Combining Transformations
KEY: translation

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2. Determine the equation, in standard form, of each parabola after being transformed from $f(x) = x^2$ by the given translations.
a) 4 units to the right and 3 units up
b) 2 units to the left and 1 unit up
c) 2 units down and 7 units to the left

ANS:

a) $g(x) = (x-4)^2 + 3$

$= x^2 - 8x + 16 + 3$

$= x^2 - 8x + 19$

b) $g(x) = (x+2)^2 + 1$

$= x^2 + 4x + 4 + 1$

$= x^2 + 4x + 5$

c) $g(x) = (x+7)^2 - 2$

$= x^2 + 14x + 49 - 2$

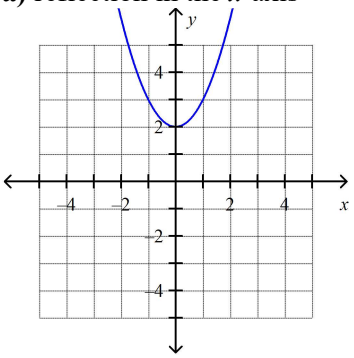
$= x^2 + 14x + 47$

DIF: Average OBJ: Section 1.3 NAT: RF4 TOP: Combining Transformations
KEY: translation

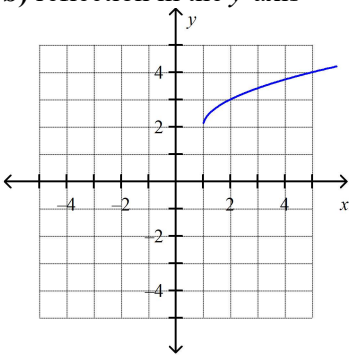
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3. Given the graph of a function, sketch the resulting graph after the specified transformation.

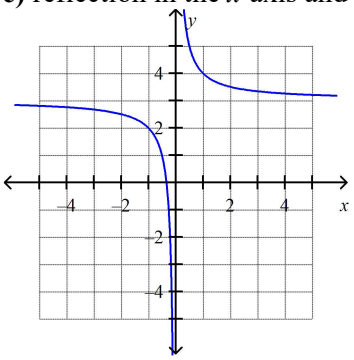
a) reflection in the *x*-axis



b) reflection in the *y*-axis

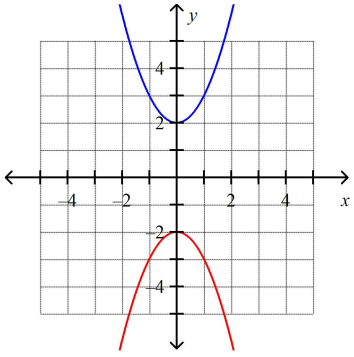


c) reflection in the *x*-axis and the *y*-axis



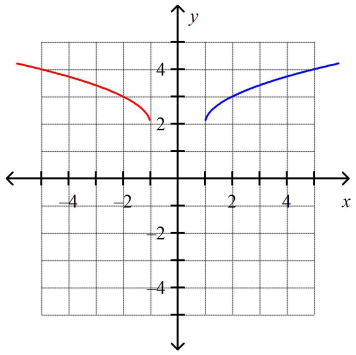
ANS:

a)

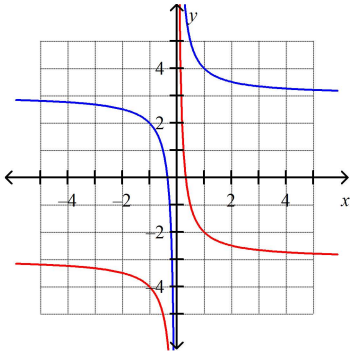


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b)



c)



DIF: Difficult + OBJ: Section 1.2 NAT: RF3 | RF5 TOP: Reflections and Stretches
KEY: graph | reflection

4. Determine the equation of the function $g(x)$ after the indicated reflection.

a) $f(x) = (x-1)^2 + 2$, in the x-axis

b) $f(x) = |x| + 1$, in the y-axis

ANS:

a) $g(x) = -f(x)$

$= -(x-1)^2 - 2$

b) $g(x) = f(-x)$

$= |-x| + 1$

$= |x| + 1$

DIF: Average OBJ: Section 1.3 NAT: RF4 | RF5 TOP: Combining Transformations
KEY: reflection | translation

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5. **a)** Sketch the graph of $g(x) = 2f(2x)$ for each base function.

i) $f(x) = x$

ii) $f(x) = x^2$

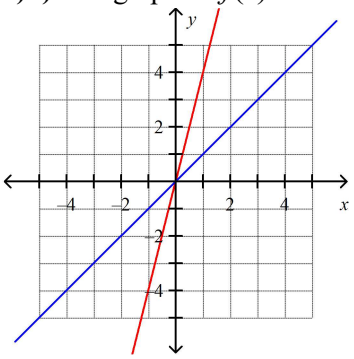
iii) $f(x) = |x|$

b) Write the equation for $g(x)$ to represent a single stretch that results in the same graph as in each function in part a).

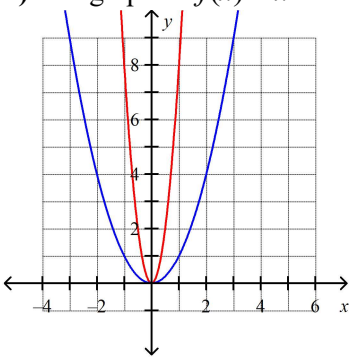
c) Describe how each stretch affects the domain and range for each function.

ANS:

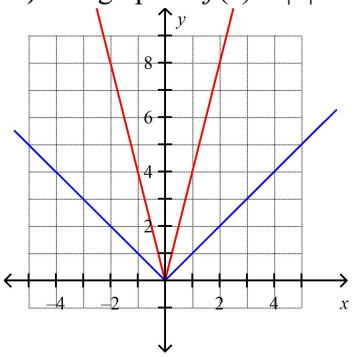
a) i) The graph of $f(x) = x$ is shown in blue and the graph of $g(x) = 2(2x)$ is shown in red.



ii) The graph of $f(x) = x^2$ is shown in blue and the graph of $g(x) = 2(2x)^2$ is shown in red.



iii) The graph of $f(x) = |x|$ is shown in blue and the graph of $g(x) = 2|2x|$ is shown in red.



b) i) $g(x) = 4x$ (vertical stretch by a factor of 4)

ii) $g(x) = 8x^2$ (vertical stretch by a factor of 8)

iii) $g(x) = 4|x|$ (vertical stretch by a factor of 4)

c) The stretches do not affect the domain or range of any of the functions.

DIF: Average OBJ: Section 1.2 NAT: RF3 TOP: Reflections and Stretches
KEY: stretch | graph

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6. For each $g(x)$, describe the transformation(s) from the base function $f(x) = x^2$.

a) $g(x) = \frac{1}{2}x^2$

b) $g(x) = \left(\frac{x}{3}\right)^2$

c) $g(x) = 5\left(\frac{x}{2}\right)^2$

ANS:

a) a vertical stretch by a factor of $\frac{1}{2}$

b) a horizontal stretch by a factor of 3

c) a vertical stretch by a factor of 5 and a horizontal stretch by a factor of 2

DIF: Average OBJ: Section 1.2 NAT: RF3 TOP: Reflections and Stretches
KEY: stretch

7. For each $g(x)$, describe, in the appropriate order, the combination of transformations that must be applied to the base function $f(x) = \sqrt{x}$.

a) $g(x) = -\sqrt{2(x+1)} - 2$

b) $g(x) = 2\sqrt{x-3} - 4$

c) $g(x) = -\frac{1}{2}\sqrt{5-x} + 1$

ANS:

a) a reflection in the x-axis, a horizontal compression by a factor of $\frac{1}{2}$, and then a translation of 1 unit to the left and 2 units down

b) a vertical stretch by a factor of 2, and then a translation of 3 units to the right and 4 units down

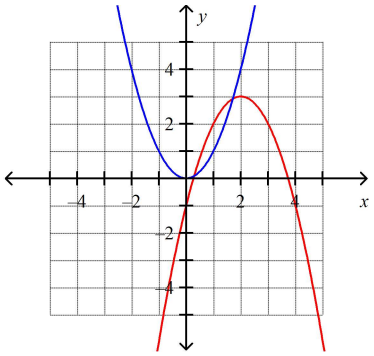
c) reflections in the x-axis and the y-axis, a vertical compression by a factor of $\frac{1}{2}$, and then a translation of 5 units to the right and 1 unit up

DIF: Difficult + OBJ: Section 1.3 NAT: RF4 | RF5 TOP: Combining Transformations
KEY: translation | stretch | reflection

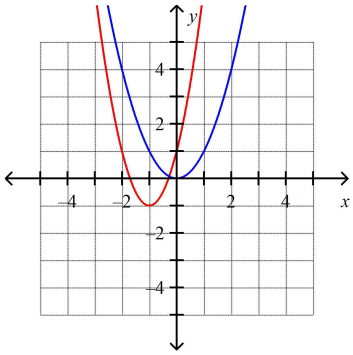
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8. For each of the following, describe the combination of transformations that must be applied to the graph of $f(x) = x^2$ (shown in blue) to obtain the graph of $g(x)$ (shown in red).

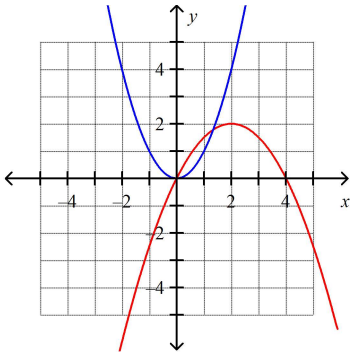
a)



b)



c)



ANS:
a) a reflection in the x-axis, and then a translation of 2 units to the right and 3 units up
b) a vertical stretch by a factor of 2, and then a translation of 1 unit to the left and 1 unit down
c) a vertical compression by a factor of $\frac{1}{2}$, a reflection in the x-axis, and then a translation of 2 units to the right and 2 units up

DIF: Average OBJ: Section 1.3 NAT: RF4 TOP: Combining Transformations
KEY: graph | transformation

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9. For each function $f(x)$,
i) determine $f^{-1}(x)$
ii) graph $f(x)$ and its inverse

a) $f(x) = \frac{5}{2}x - 3$

b) $f(x) = 3(x-2)^2 - 3$

ANS:

a) i) $y = \frac{5}{2}x - 3$

$x = \frac{5}{2}y - 3$

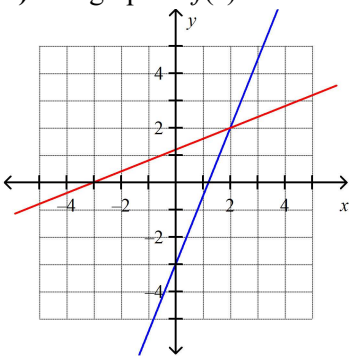
$x + 3 = \frac{5}{2}y$

$2x + 6 = 5y$

$y = \frac{2}{5}x + \frac{6}{5}$

$f^{-1}(x) = \frac{2}{5}x + \frac{6}{5}$

ii) The graph of $f(x)$ is shown in blue and the graph of $f^{-1}(x)$ is shown in red.



b) i) $y = 3(x-2)^2 - 3$

$x = 3(y-2)^2 - 3$

$x + 3 = 3(y-2)^2$

$\frac{x+3}{3} = (y-2)^2$

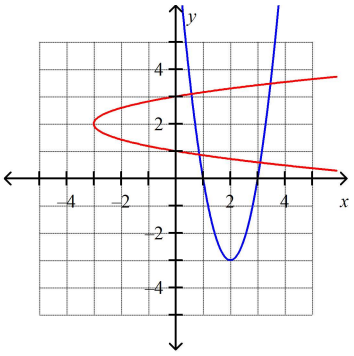
$\pm\sqrt{\frac{x+3}{3}} = y - 2$

$y = 2 \pm \sqrt{\frac{x+3}{3}}$

$f^{-1}(x) = 2 \pm \sqrt{\frac{x+3}{3}}$

ii) The graph of $f(x)$ is shown in blue and the graph of $f^{-1}(x)$ is shown in red.

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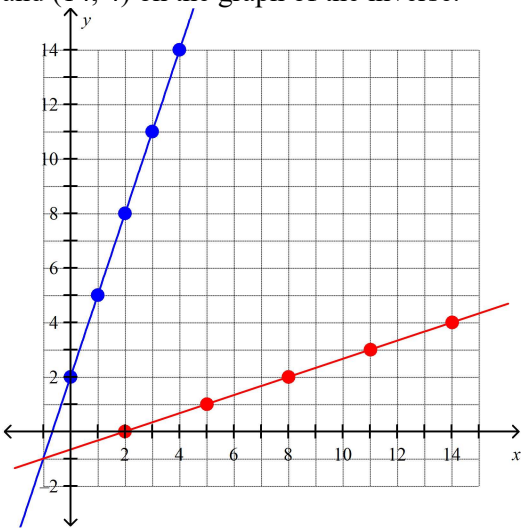


DIF: Difficult OBJ: Section 1.4 NAT: RF6 TOP: Inverse of a Relation
KEY: inverse of a function | graph | function notation

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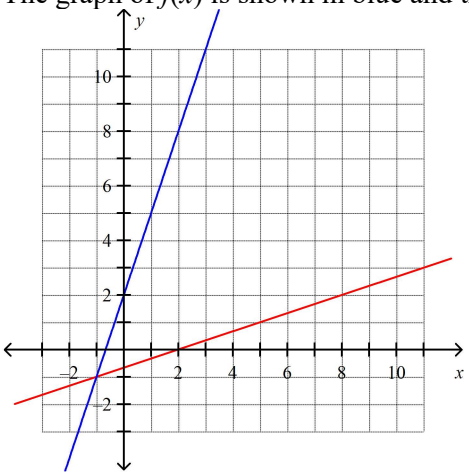
10. **a)** Determine the inverse graph of the function $f(x) = 3x + 2$ in two ways:
i) Determine 5 points on the line, switch their x-coordinates and y-coordinates, and graph the new set of points, connecting these new points with a straight line.
ii) Algebraically determine the inverse of the function and graph this function.
b) How do your answers in part a) compare?

ANS:
a) i) Answers may vary. Sample answer:
Points (0, 2), (1, 5), (2, 8), (3, 11), and (4, 14) on the graph of $f(x)$ become (2, 0), (5, 1), (8, 2), (11, 3), and (14, 4) on the graph of the inverse.



ii)
$$y = 3x + 2$$
$$x = 3y + 2$$
$$x - 2 = 3y$$
$$\frac{1}{3}x - \frac{2}{3} = y$$
$$f^{-1}(x) = \frac{1}{3}x - \frac{2}{3}$$

The graph of $f(x)$ is shown in blue and the graph of its inverse is shown in red.



b) Answers may vary. Sample answer:
Each point graphed in part a) i) is on the corresponding line graphed in ii).

DIF: Difficult OBJ: Section 1.4 NAT: RF6 TOP: Inverse of a Relation

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KEY: graph | inverse of a function | function notation

11. **a)** Determine the inverse of the function $f(x) = x^2 - 4x + 5$.
b) Determine the domain and range of the function and its inverse.
c) Is the inverse a function? Explain.

ANS:
a) First, rewrite the quadratic function in vertex form.

$$f(x) = x^2 - 4x + 5$$
$$= (x^2 - 4x + 4 - 4) + 5$$
$$= (x-2)^2 + 1$$

Then, determine the inverse.

$$x = (y-2)^2 + 1$$
$$x - 1 = (y-2)^2$$
$$\pm\sqrt{x-1} = y - 2$$
$$2 \pm \sqrt{x-1} = y$$
$$f^{-1}(x) = 2 \pm \sqrt{x-1}$$

b) For $f(x)$: domain $\{x \in \mathrm{R}\}$, range $\{y \geq 1, y \in \mathrm{R}\}$
For $f^{-1}(x)$: domain $\{x \geq 1, x \in \mathrm{R}\}$, range $\{y \in \mathrm{R}\}$
c) The inverse is not a function because it fails the vertical line test.

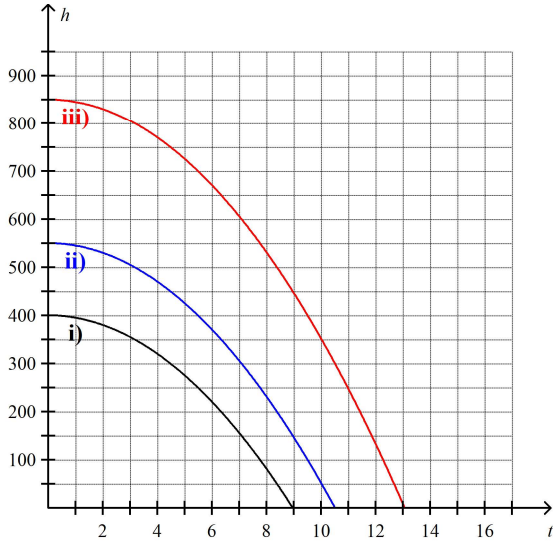
DIF: Average OBJ: Section 1.4 NAT: RF6 TOP: Inverse of a Relation
KEY: inverse of a function | domain | range | graph | function notation | completing the square

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PROBLEM

1. The approximate height, h, in metres, of an object above the ground after it is dropped from a distance, d, in metres, can be modelled by the function $h(t) = d - 5t^2$, where t is the time, in seconds, after being dropped.
a) Determine the function for an object being dropped from each height.
i) 400 m
ii) 550 m
iii) 850 m
b) Graph all three functions on the same set of axes.
c) Describe the transformation that relates the highest curve to the lowest curve.
d) Describe the transformation that relates the highest curve to the curve in the middle.

ANS:
a) i) $h_{400}(t) = 400 - 5t^2$
ii) $h_{550}(t) = 550 - 5t^2$
iii) $h_{850}(t) = 850 - 5t^2$
b)



c) a vertical translation of 450 units up; $h_{850}(t) = h_{400}(t) + 450$
d) a vertical translation of 300 units up; $h_{850}(t) = h_{550}(t) + 300$

DIF: Easy OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations KEY: vertical translation

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2. To rent a cottage for the months of July and August, the renter must pay a flat fee of $150 plus a daily amount of $100.
a) Write a function that represents the total cost, C, in dollars, of the rental as a function of the number of days, d, that the cottage is rented.
b) Determine the domain and range of the function.
c) In the months of November and December, the renter only needs to pay a flat fee of $50, but the daily fee is still $100. Write a function that represents the total cost, T, in dollars, as a function of the number of days, d, of the rental.
d) Describe a transformation that relates the function $C(d)$ to the function $T(d)$.

ANS:
a) $C(d) = 100d + 150$
b) domain $\{0 < d \leq 62, d \in \mathrm{N}\}$, range $\{0 < C \leq 6350, C \in \mathrm{N}\}$
c) $T(d) = 100d + 50$
d) The graph of $C(d)$ is a translation of 100 units up of the graph of $T(d)$; $C(d) = T(d) + 100$.

DIF: Average OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations
KEY: translation | function notation

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3. Consider the function $f(x) = 2(x-1)^2 - 3$.

a) Determine the equation of each function.

i) $-f(x)$

ii) $f(-x)$

iii) $-f(-x)$

b) Graph all four functions from part a) on the same set of axes.

c) From the graph, determine the pairs of equations that can be represented as translations of each other.

d) Describe the translation that can be applied to each pair of functions you determined in part c) to generate the same graph.

ANS:

a) i) $-f(x) = -[2(x-1)^2 - 3]$

$= -2(x-1)^2 + 3$

ii) $f(-x) = 2(-x-1)^2 - 3$

$= 2(-1)^2(x+1)^2 - 3$

$= 2(x+1)^2 - 3$

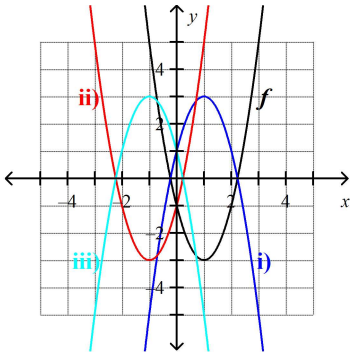
iii) $-f(-x) = -[2(-x-1)^2 - 3]$

$= -[2(-1)^2(x+1)^2 - 3]$

$= -[2(x+1)^2 - 3]$

$= -2(x+1)^2 + 3$

b)



c) The graphs of $f(x)$ and $f(-x)$ are horizontal translations of each other. This is also true of $-f(x)$ and $-f(-x)$.

d) In both pairs, one curve is a horizontal translation of 2 units left or right of the other.

DIF: Difficult OBJ: Section 1.3 NAT: RF4 TOP: Combining Transformations
KEY: reflection | graph | translation | function notation

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4. For the base function $f(x) = x^2$, a reflection in the y-axis is followed by a reflection in the x-axis.

a) Discuss the effect of each reflection on the base function.

b) Is there a point that you expect to be invariant during these reflections? If so, state its coordinates.

c) Graph the function after each reflection.

d) Use your graph in part c) to verify your answer for part b).

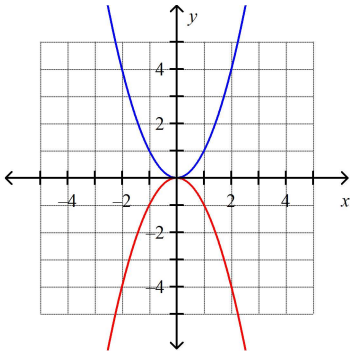
e) Make a general statement regarding invariant points after reflections in the x-axis and the y-axis.

ANS:

a) The reflection in the y-axis does not change the graph of the function. The reflection in the x-axis does change the graph, so that the graph of the function points downward.

b) Since the vertex is on the x-axis and the y-axis, at (0, 0), this point should not change during the reflections.

c)



d) The graph shows that the point (0, 0) is invariant under the reflections.

e) Invariant points after a reflection in the x-axis are those points of the original function that are on the x-axis, and invariant points after a reflection in the y-axis are those points of the original function that are on the y-axis. The only invariant point after both reflections is the origin.

DIF: Average OBJ: Section 1.2 NAT: RF3 TOP: Reflections and Stretches
KEY: graph | reflection | invariant point

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5. For the base function $f(x) = |x|$, a reflection in the y-axis is followed by a reflection in the x-axis.

a) Discuss the effect of each reflection on the base function.

b) Is there a point that you expect to be invariant during these reflections? If so, state its coordinates.

c) Graph each function after the reflections.

d) Use this graph to verify your answer for part b).

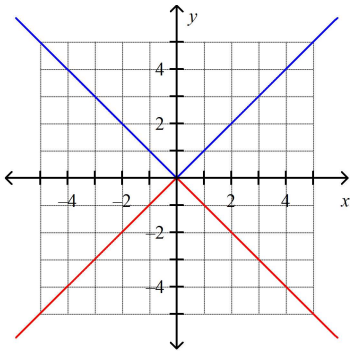
e) Make a general statement regarding invariant points after reflections in the x-axis and the y-axis.

ANS:

a) The reflection in the y-axis does not change the graph of the function. The reflection in the x-axis does change the graph of the function so that it points downward after the reflection.

b) Since the vertex is on the x-axis and the y-axis, at (0, 0), this point should not change during the reflections.

c)



d) The graph shows that the point (0, 0) is invariant under the reflections.

e) Invariant points after a reflection in the x-axis are those points of the original function that are on the x-axis, and invariant points after a reflection in the y-axis are those points of the original function that are on the y-axis. The only invariant point after both reflections is the origin.

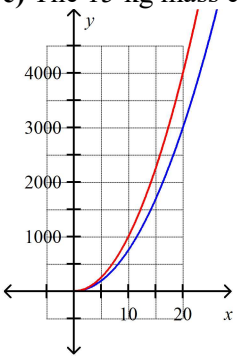
DIF: Average OBJ: Section 1.2 NAT: RF3 TOP: Reflections and Stretches
KEY: graph | reflection | invariant point

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6. The kinetic energy (energy of motion), E, in joules, of an object is given by the equation $E = \frac{1}{2}mv^2$, where m represents the mass of the object, in kilograms, and v represents its speed, in metres per second.

a) Determine the equation for the kinetic energy of a mass of 15 kg as a function of its speed.
b) Find the kinetic energy of an object of mass 15 kg moving at
i) 5 m/s
ii) 10 m/s
iii) 15 m/s
iv) 20 m/s
c) Determine the equation for the kinetic energy of a mass of 20 kg as a function of its speed.
d) Find the kinetic energy of an object of mass 20 kg moving at
i) 5 m/s
ii) 10 m/s
iii) 15 m/s
iv) 20 m/s
e) Graph the functions in parts b) and d) on the same set of axes.
f) Explain how the two functions are vertical stretches of each other, and determine the value of the vertical stretch factor.

ANS:
a) $E = 7.5v^2$
b) i) 187.5 J **ii)** 750 J **iii)** 1687.5 J **iv)** 3000 J
c) $E = 10v^2$
d) i) 250 J **ii)** 1000 J **iii)** 2250 J **iv)** 4000 J
e) The 15-kg mass curve is shown in blue and the 20-kg mass curve is shown in red.



f) Since the two functions differ in their value of a in $E = av^2$, they are vertical stretches of each other. The value of this vertical stretch is $\frac{4}{3}$ for the function describing the 20-kg object compared to the function describing the 15-kg object.

$\frac{4}{3}(7.5) = 10$

DIF: Difficult + OBJ: Section 1.2 NAT: RF3 TOP: Reflections and Stretches
KEY: graph | stretch | function notation

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7. An object falls to the ground from a height of 25 m. The height, h, in metres, of the object above the ground can be modelled by the function $h(t) = -\frac{1}{2}at^2 + 25$, where a is the acceleration due to gravity, in metres per second squared, and t is the time, in seconds.

a) Write an equation for the height of the object on Earth given $a = 9.8$ m/s^2.

b) Write an equation for the height of the object on Mars given $a = 3.7$ m/s^2.

c) Graph both functions on the same set of axes.

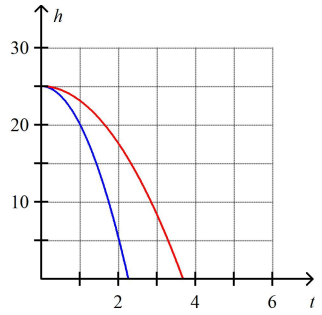
d) What scale factor can be applied to the Earth function to transform it to the Mars function?

ANS:

a) $h(t) = -4.9t^2 + 25$

b) $h(t) = -1.85t^2 + 25$

c) The Earth function is shown in blue and the Mars function is shown in red.



d) The scale factor that can be applied to the Earth function to transform it to the Mars function is $\frac{37}{98}$.

$4.9 \times \frac{37}{98} = 1.85$

DIF: Difficult OBJ: Section 1.2 NAT: RF3 TOP: Reflections and Stretches
KEY: graph | stretch | function notation

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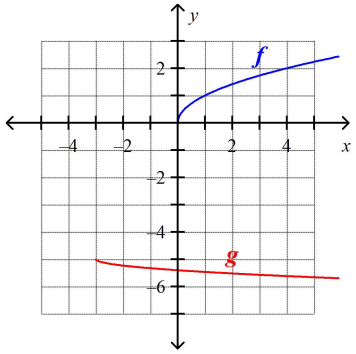
8. The base function $f(x) = \sqrt{x}$ is reflected in the x-axis, stretched horizontally by a factor of 2, compressed vertically by a factor of $\frac{1}{3}$, and translated 3 units to the left and 5 units down.

a) Write the equation of the transformed function $g(x)$.
b) Graph the original function and the transformed function on the same set of axes.
c) Which transformations must be done first but in any order?
d) Which transformations must be done last but in any order?

ANS:

a) $g(x) = -\frac{1}{3}\sqrt{\frac{1}{2}(x+3)} - 5$

b)



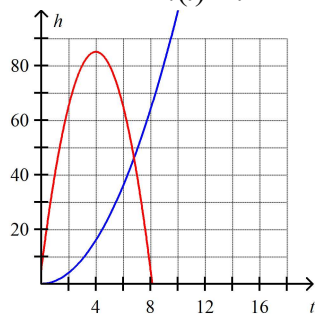
c) The reflection, horizontal stretch, and vertical compression must be done first, but can be done in any order.
d) The translations to the left and down must be done last, but can be done in any order.

DIF: Difficult + OBJ: Section 1.3 NAT: RF4 TOP: Combining Transformations
KEY: graph | transformation | function notation

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9. The height, h, in metres, of a flare fired into the air from a boat can be modelled using the function $h(t) = -5(t-4)^2 + 85$, where t is the time, in seconds, after the flare is fired.
a) Explain how this function can be obtained from the function $h(t) = t^2$ using a series of transformations.
b) Graph the base function and the transformed function on the same set of axes.
c) Use the graph to estimate
i) how long the flare is in the air
ii) the maximum height reached by the flare, and how long it takes to reach this maximum height

ANS:
a) The graph of the function $h(t) = t^2$ must be reflected in the t-axis, vertically stretched by a factor of 5, and then translated 4 units to the right and 85 units up.
b) The function $h(t) = t^2$ is shown in blue and the function $h(t) = -5(t-4)^2 + 85$ is shown in red.



c) i) The flare is in the air for approximately 8 s.
ii) The flare reaches a maximum height of 85 m after 4 s.

DIF: Average OBJ: Section 1.3 NAT: RF4 | RF5 TOP: Combining Transformations
KEY: graph | transformation

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10. To graph the inverse of a linear function, you can use the following method.

1. Graph the linear function and the line $y = x$ on grid paper using the same scale for both axes.

2. Draw light pencil lines that are perpendicular to the line $y = x$. Measure the distances along these lines from the line $y = x$ to the linear function.

3. Mark these measurements along the perpendicular lines on the opposite side of the line $y = x$.

4. Connect these points to form the graph of the inverse of the function.

a) Use this method to graph the function $f(x) = 2x + 4$ and its inverse.

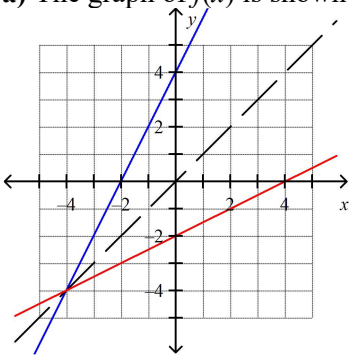
b) Determine the inverse function $f^{-1}(x)$ algebraically.

c) Graph $f^{-1}(x)$ on the same axes as you used in part a) to verify the result.

d) Is there a point that is invariant on the graph? If so, what are the coordinates of the point?

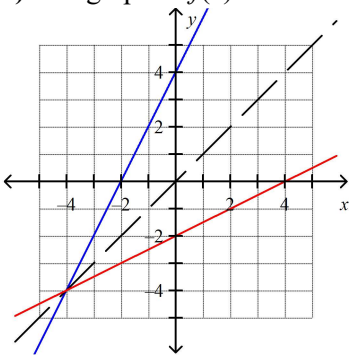
ANS:

a) The graph of $f(x)$ is shown in blue and the graph of its inverse is shown in red.



b)
$$y = 2x + 4$$
$$x = 2y + 4$$
$$x - 4 = 2y$$
$$y = \frac{1}{2}x - 2$$
$$f^{-1}(x) = \frac{1}{2}x - 2$$

c) The graph of $f(x)$ is shown in blue and the graph of $f^{-1}(x)$ is shown in red.



d) The invariant point is (–4, –4).

DIF: Easy OBJ: Section 1.4 NAT: RF6 TOP: Inverse of a Relation
KEY: graph | inverse of a function | function notation

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11. Consider the function $f(x) = -2\sqrt{x+1} - 4$.

a) State the domain and range of the function.

b) Use this information to determine the domain and range of the inverse of the function.

c) Determine the inverse of the function.

d) Graph the function and its inverse on the same set of axes (include the line $y = x$ to verify the inverse).

ANS:

a) domain $\{x \geq -1, x \in \mathrm{R}\}$, range $\{y \leq -4, y \in \mathrm{R}\}$

b) domain $\{x \leq -4, x \in \mathrm{R}\}$, range $\{y \geq -1, y \in \mathrm{R}\}$

c)

$$y = -2\sqrt{x+1} - 4$$

$$x = -2\sqrt{y+1} - 4$$

$$x + 4 = -2\sqrt{y+1}$$

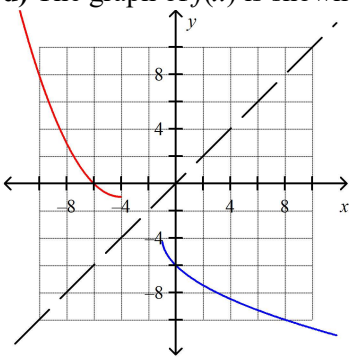
$$-\frac{(x+4)}{2} = \sqrt{y+1}$$

$$\left(-\left(\frac{x+4}{2}\right)\right)^2 = y + 1$$

$$y = \left(\frac{x+4}{2}\right)^2 - 1$$

$$f^{-1}(x) = \frac{1}{4}(x+4)^2 - 1$$

d) The graph of $f(x)$ is shown in blue and the graph of $f^{-1}(x)$ is shown in red.



DIF: Difficult OBJ: Section 1.4 NAT: RF6 TOP: Inverse of a Relation
KEY: graph | inverse of a function | function notation | domain | range

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12. The cost of renting a car for a day is a flat fee of $50 plus $0.12 for each kilometre driven. Let C represent the total cost of renting a car for a day if it is driven a distance, x, in kilometres.
a) Write the total cost function for the car rental.
b) Determine the inverse of this function.
c) What does this inverse function represent?
d) Give an example of how this function can be used.

ANS:
a) $C(x) = 50 + 0.12x$
b) $C = 50 + 0.12x$

$$C - 50 = 0.12x$$

$$x = \frac{C - 50}{0.12}$$

$$f^{-1}(C) = \frac{C - 50}{0.12}$$

c) This function represents the distance the car can be driven for a given rental cost.
d) Answers may vary. Sample answer:
If you have $65 to spend on the car rental, for how many kilometres can you drive the car?

$$f^{-1}(65) = \frac{65 - 50}{0.12}$$

$$= 125$$

You can drive a total of 125 km for the $65 rental fee.

DIF: Average OBJ: Section 1.4 NAT: RF6 TOP: Inverse of a Relation
KEY: inverse of a function | function notation

13. The graph of $f(x) = x^2$ is transformed to the graph of $g(x) = f(x + 8) + 12$.
a) Describe the two translations represented by this transformation.
b) Determine three points on the base function. Horizontally translate and then vertically translate the points. What are the three resulting image points?

ANS:
a) The graph is translated 8 units to the left and 12 units upward.
b) Answers may vary. Sample answer:
Start with the points (0, 0), (1, 1), and (−1, 1). Translating the points 8 units to the left results in the coordinates (−8, 0), (−7, 1), and (−9, 1). Translating the points 12 units upward changes the coordinates to (−8, 12), (−7, 13), and (−9, 13).

DIF: Easy OBJ: Section 1.1 NAT: RF2
TOP: Horizontal and Vertical Translations
KEY: horizontal translation | vertical translation

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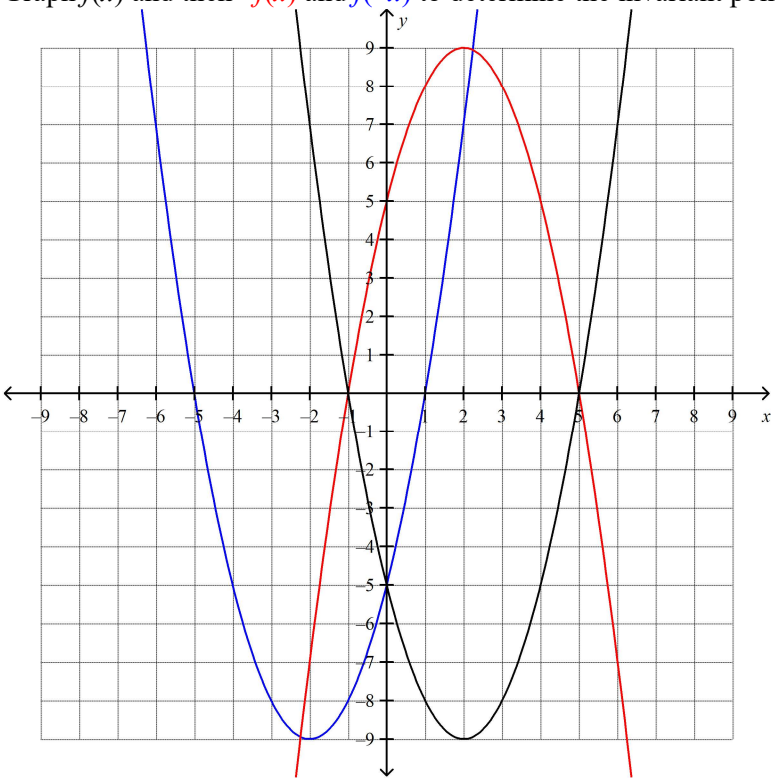
14. Draw a graph to determine the invariant points of $f(x) = (x-2)^2 - 9$ when it is reflected in the

a) x-axis

b) y-axis

ANS:

Graph $f(x)$ and then $-f(x)$ and $f(-x)$ to determine the invariant points.



a) From the graph, the invariant points are $(-1, 0)$ and $(5, 0)$.

b) From the graph, the invariant point is $(0, -5)$.

DIF: Easy OBJ: Section 1.2 NAT: RF3 TOP: Reflections and Stretches

KEY: invariant point | graph

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15. Two skydivers jump out of a plane. The first skydiver's motion can be modelled by the function $g(t) = 4000 - 5(t + 10)^2$. The second skydiver jumps 10 s later with a goal of catching up to the first skydiver. The motion of the second skydiver can be modelled by $h(t) = 4000 - 5t^2$. For both functions, the height is measured in metres and the time is the number of seconds after the second skydiver jumps.

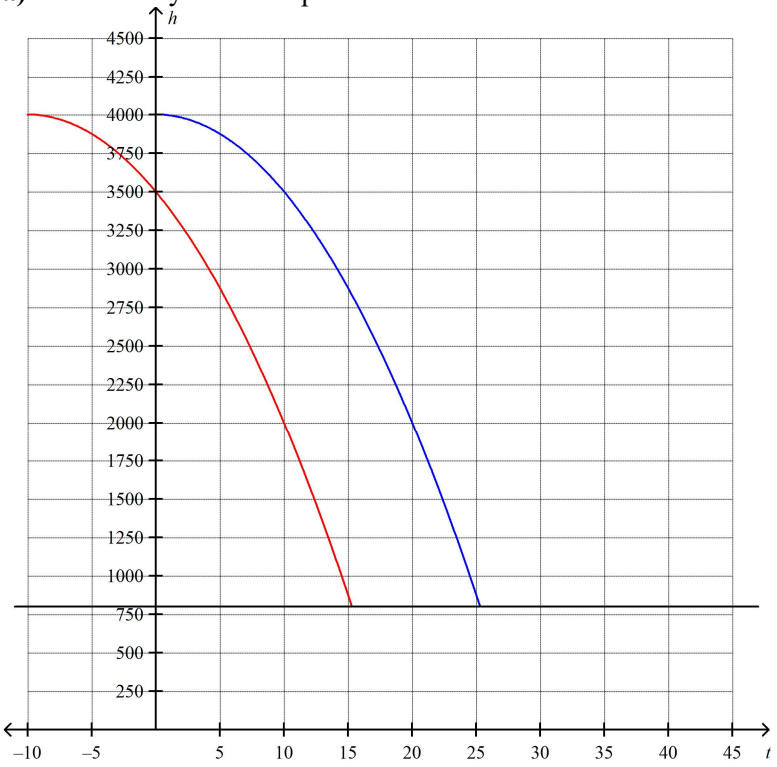
a) Graph the functions on the same set of axes.

b) Will the second skydiver catch up to the first before they have to open their parachutes at 800 m?

c) State the domain and range of these functions in this context.

ANS:

a) The first skydiver is represented in red and the second in blue.



b) No; from the graphs it can be seen that the path of the second skydiver will not intersect the path of the first before they reach 800 m above the ground.

c) The domain for the first skydiver is $\{-10 \le t \le 15.3, t \in \mathrm{R}\}$, while for the second skydiver it is $\{0 \le t \le 25.3, t \in \mathrm{R}\}$. The range for $g(t)$ is $\{800 \le g \le 4000, g \in \mathrm{R}\}$, and the range for $h(t)$ is $\{800 \le h \le 4000, h \in \mathrm{R}\}$. The given functions do not apply after the skydivers pass 800 m above the ground, because when they open their parachutes, the function modelling their height changes.

DIF: Average OBJ: Section 1.3 NAT: RF4 TOP: Combining Transformations
KEY: translation | graph

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16. The value, V, in thousands of dollars, of a certain car after t years can be modelled by the equation $V(t) = \frac{35}{t+3}$.

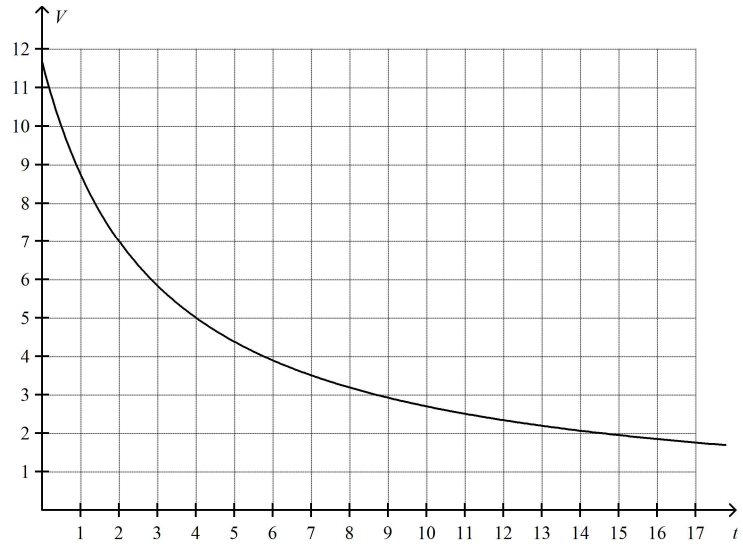
a) Sketch the graph of this relation using transformations of the function $f(t) = \frac{1}{t}$.

b) What is the initial value of this car?

c) What is the projected value of this car after 2 years?

ANS:

a)



b) From the graph, the y-intercept is about 11.5, so the initial value of the car is about \$11 500.

c) At $t = 2$, the value of the car is \$7000.

DIF: Difficult + OBJ: Section 1.3 NAT: RF4 TOP: Combining Transformations
KEY: stretch | translation | graph

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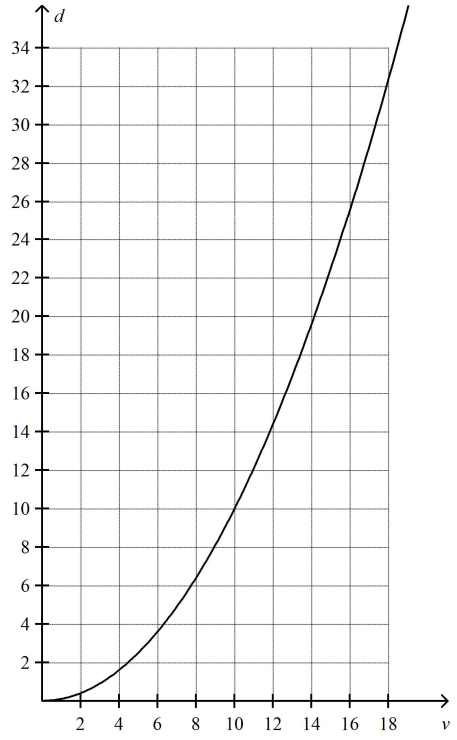
17. In ballistics, the paths of projectiles are studied. It is known that the greater the speed at which an object is fired, the farther it will travel. For a particular spring-loaded cannon, this relationship can be modelled by $d = \frac{v^2}{10}$, where d is the distance, in metres, travelled by the projectile and v is the muzzle speed, in metres per second.

a) Graph the distance function and state its domain and range.

b) Graph the inverse of the distance function. What does it represent? State its domain and range.

ANS:

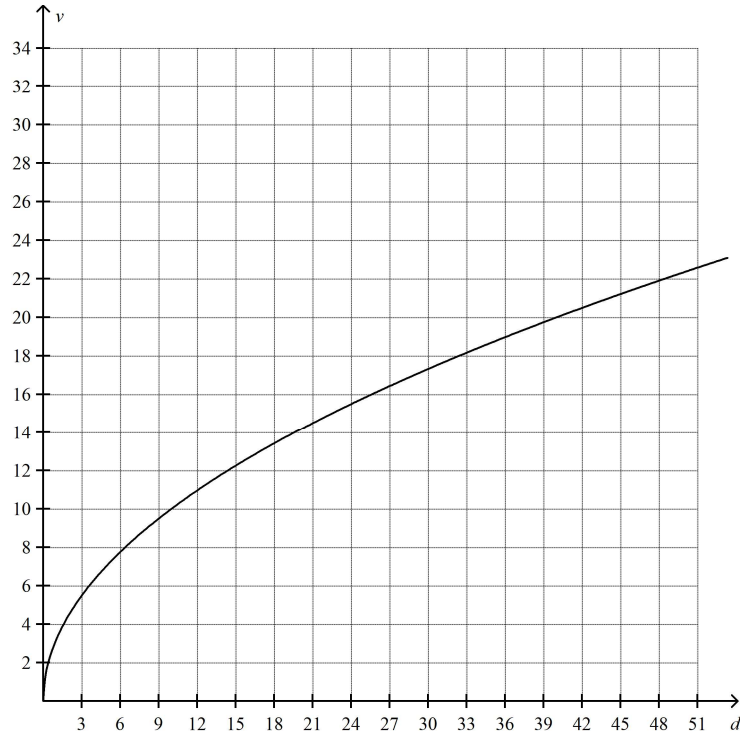
a)



The domain is $\{v \geq 0, v \in \mathrm{R}\}$ and the range is $\{d \geq 0, d \in \mathrm{R}\}$

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b)



This graph represent the velocity required to reach a certain distance. The domain is $\{d \geq 0, d \in \mathrm{R}\}$ and the range is $\{v \geq 0, v \in \mathrm{R}\}$.

DIF: Difficult OBJ: Section 1.4 NAT: RF6 TOP: Inverse of a Relation
KEY: inverse of a function

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18. The equation $y = \frac{9}{5}x + 32$ can be used to convert between Celsius and Fahrenheit temperatures, where x is the temperature in degrees Celsius and y is the temperature in degrees Fahrenheit.

a) Determine the inverse of this equation. What does it represent? What do the variables represent?

b) Graph the original and inverse functions on the same set of axes.

c) Which temperature is the same in Celsius and Fahrenheit? Explain how you know.

ANS:

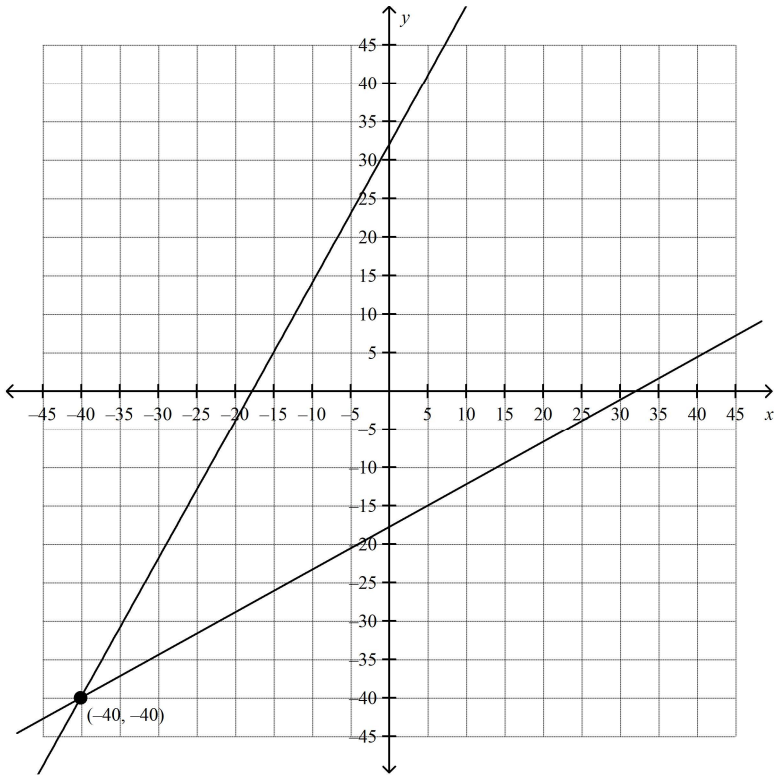
a) $x = \frac{9}{5}y + 32$

$x - 32 = \frac{9}{5}y$

$y = \frac{5}{9}(x - 32)$

The inverse represents the equation to convert a temperature from degrees Fahrenheit to degrees Celsius. The variable x represents the temperature in degrees Fahrenheit and the variable y represents the temperature in degrees Celsius.

b)



c) The temperature that is the same in Celsius and Fahrenheit is –40°. This is the invariant point of the original function and its inverse.

DIF: Average OBJ: Section 1.4 NAT: RF6 TOP: Inverse of a Relation
KEY: inverse of a function | invariant point

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19. Jai works at an electronics store. She earns $600 a week, plus commission of 5% of her sales.
a) Write a function to describe Jai's total weekly earnings as a function of her sales.
b) Determine the inverse of this function.
c) One week Jai earned $775. What were her sales that week?

ANS:
a) Let s represent Jai's sales that week and t her total earnings.
$t = 600 + 0.05s$
b) $t = 600 + 0.05s$

$t - 600 = 0.05s$

$$s = \frac{t - 600}{0.05}$$

$s = 20t - 12\,000$
c) $s = 20(775) - 12\,000$

$= 3500$
Jai had sales of $3500 that week

DIF: Average OBJ: Section 1.4 NAT: RF6 TOP: Inverse of a Relation
KEY: inverse of a function

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20. Given the function $f(x) = 3x + 7$,

a) determine the inverse of the function

b) graph the function and its inverse on the same set of axes

ANS:

a) $f(x) = 3x + 7$

$$y = 3x + 7$$

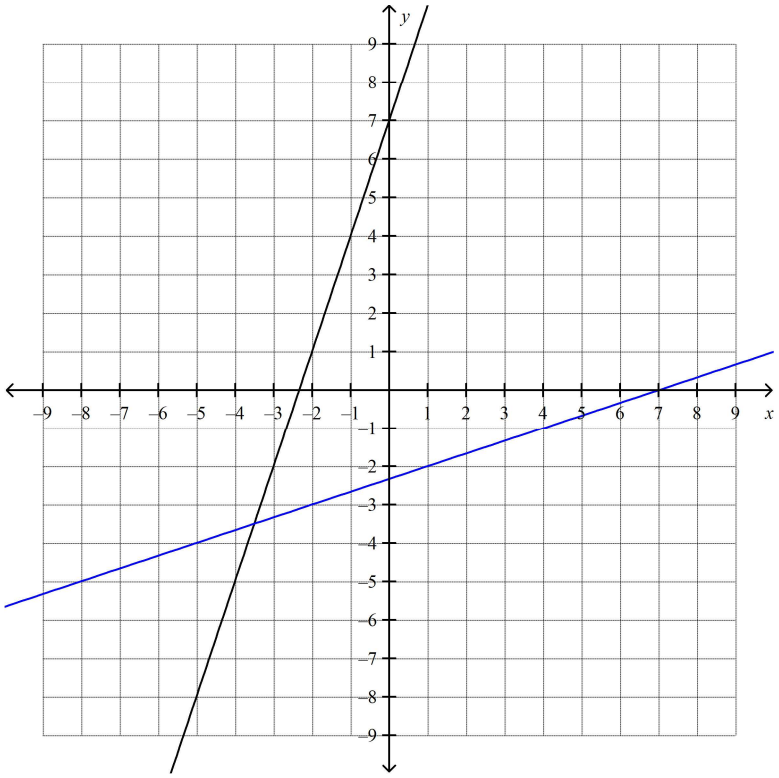
$$x = 3y + 7$$

$$x - 7 = 3y$$

$$y = \frac{x - 7}{3}$$

$$f^{-1}(x) = \frac{x - 7}{3}$$

b)



DIF: Easy OBJ: Section 1.4 NAT: RF6 TOP: Inverse of a Relation
KEY: inverse of a function | graph

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